MindFrame: The Whole Model as One Equation

The master function, the Kai shield, and a faith reading

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MindFrame is a model built from design assumptions and informed by research. It is not validated science, not a diagnosis, and not therapy. In a crisis in the US, call or text 988, or call 911.

All credit and glory to God. I owe my knowledge to my Lord Jesus Christ.

1 The whole model on one page

MindFrame reads one thought into a vector $e \in [0,1]^9$ of nine movers (fear, sadness, anger, shame, calm, connection, hope, rumination, avoidance) and takes one gentle step toward a home state h. Every feature built so far fits in four lines:

$$e = \mathrm{clip}_{[0,1]}\big(e_- \oplus n(\text{text}) + X\,(1-S)\,w\big) \qquad \text{the reading} \tag{1}$$

$$e' = \mathrm{clip}_{[0,1]}\big(e + a(e)\,(h-e) + b\,s(e)\big) \qquad \text{the step} \tag{2}$$

$$P = P_d(e)\,(1-\gamma C)\,(1-\delta G);\quad \ell = \arg\max_i \Delta P_i;\quad \text{step} = \text{practice}(\ell) \qquad \text{peace} \tag{3}$$

$$\text{words} = \text{compose}(e, e', P, \ell, \text{passes}). \tag{4}$$

- e_- is where the person was: the session's e, or the vector picked with the feeling chips. $n(\text{text})$ is what the words add: a lexicon, thinking-trap signals, and word hooks (relationship patterns, forgiveness cues, true size, patience, body tension), each saturating as $1 - e^{-\text{raw}}$. The saturating sum $\oplus$ is $1-(1-e_j)(1-n_j)$ for hard and process movers and $e_j + n_j^+(1-e_j) - n_j^- e_j$ for the grounding movers (calm, connection, hope).
- $X(1-S)w$ is the Kai term (Section 2). For almost every text $X = 0$, and line (1) is just $e_- \oplus n$.
- $a(e) = \min(a_{\max}, a_0 + k\|e-h\|)$, lowered to $a_{\text{cap}} = 0.12$ (with $b = b_U$) when the active hard movers feed each other ($\lambda > 0$). The term $s_j(e) = e_j \sum_g \min(0, W_{gj})\,h_g \le 0$ is grounding pressing on a hard or process mover that sits above home.
- $P_d = 1/(1+(d/d_0)^2)$ with $d = \|e-h\|$. C is the strongest hidden conflict $\sqrt{x_i y_j}$. $G = \max(g,u)$ is the true-size signal (too big, g; too small, u).
- ΔP_i is the gain in P when e_i moves 0.05 toward h_i. The lever ℓ is the largest gain. practice(ℓ) is a fixed, deterministic rule table: for example, the cold-water reset only for very high fear or anger with low peace, and the brain dump for very high rumination. Strictly, the step is the practice for the lever that maximizes ΔP; the model does not search every practice.
- The *passes* run in the same call and never move e. They are forgiveness ($F_{\text{self}} = e_{\text{shame}}(0.4 + 0.6c_{\text{self}})$ and $F_{\text{other}} = e_{\text{anger}}(0.4 + 0.6c_{\text{other}})$), the freeing mix ($\max J = P - \mu N$ over small band moves), the Kai lines, and the companion. The companion grows only from caring actions (an Update, a step tried, a journal save, a practice finished), never from e.
- Values are rounded to 0.01 after (1) and after (2), as the code does. The baseline $B_t = 0.8B_{t-1} + 0.2e_t$ and the recovery count (repeat (2) until $d < 0.3$) reuse the same map.

Code and formula agree. A unit test recomputes (1)–(3) from these lines alone and checks that `master()` returns exactly the same e, e', P, lever and practice. The refactor changed no other behavior: all earlier tests still pass.

2 The Kai equation: steadiness under outside pressure

In Ryan's story, Kai faces a storm of criticism. He anchors in self-awareness, applies forbearance and empathy, and his respect guard blocks the barbs without harming anyone, so impulsive reaction "flickers and fails." The model adds no movers:

$X = 1 - e^{-0.7k_X}$	hurt received (aimed *at* you)
$A = (1-G)\,[1-(1-c_A)(1-\frac{1}{2}\sigma_A)]$	self-awareness
$F = 1-(1-c_F)(1-\frac{1}{2}\sigma_F)(1-p_F)$	forbearance
$E = 1-(1-c_E)(1-\frac{1}{2}\sigma_E)$	empathy
$R = (1-c_{\text{attack}})\,[1-(1-c_R)(1-\frac{1}{2}\sigma_R)]$	respect guard
$S = \min\big(0.99,\ 1-(1-A)(1-F)(1-E)(1-R)\big)$	the shield
$e_j \leftarrow \mathrm{clip}_{[0,1]}\big(e_j + X(1-S)\,w_j\big),\quad w = (0.8, 0.6, 0.4)$ for anger, shame, fear	the push
$I = X(1-S)\,u,\quad u = \max(e_{\text{anger}}, c_{\text{urge}})$	impulsive reaction

Here each $c = 1 - e^{-0.7k}$ counts k matching phrases, with negation handled ("nobody judged me" gives $X = 0$). $\sigma = 1$ when a matching practice was finished this session: the pause or breathing practices for F, labeling for A, and the REACH "Empathize" step for E. Sessions are kept in memory only. $p_F = \frac{1}{2}$ comes from a picker chip. Each virtue is capped at 0.95. The Four Horsemen (Gottman and Levenson, 1992) in the person's *own* words never raise X; contempt or criticism aimed outward lowers R through c_{attack}.

On the card (words only, and only when $X > 0$), for the reading "My coworkers criticized me in front of everyone, I want to snap back but I know they're stressed": $X = 0.75$, $A = 0$, $F = 0.5$, $E = 0.5$, $R = 0$, $S = 0.75$, $I = 0.14$. The card says: "Your steadiness: self-awareness just starting, forbearance growing, empathy growing, respect guard just starting. The urge to react flickers and fails." The weakest virtue is the lever, and it comes with one step. For the respect guard, that step is DEAR MAN-style (Linehan, 2015): "I hear you. I see it differently, and I'm going to step away now."

Sources, honestly. Naming feelings dampens amygdala response (Lieberman et al., 2007); affirming core values buffers threat to the self (Cohen and Sherman, 2014). Patience relates to well-being and can be trained (Schnitker, 2012). Perspective taking is a distinct facet of empathy (Davis, 1983). Assertiveness training has a substantial, if older, evidence base (Speed et al., 2018). None of these studies tested this equation. The product form of S, the weights w, and the $\frac{1}{2}$ terms are modeling choices.

3 The faith reading

With *Include faith* on, the home h is described as rest and peace in Christ: "Come to me, all you who labor and are heavily burdened, and I will give you rest" (Matthew 11:28–30), and "the peace of God, which surpasses all understanding, will guard your hearts and your thoughts in Christ Jesus" (Philippians 4:7). The virtues are read as fruits of the Spirit: "love, joy, peace, patience, kindness, goodness, faith, gentleness, and self-control" (Galatians 5:22–23). Verse wording follows the World English Bible (public domain).

Model	Fruit of the Spirit	Picture in the Gospel
Forbearance F	patience	He "answered nothing" before his accusers (Matthew 27:12–14); "as a sheep that before its shearers is silent" (Isaiah 53:7)
Empathy E	kindness	"Father, forgive them, for they don't know what they are doing" (Luke 23:34)
Respect guard R	gentleness	"When he was cursed, he didn't curse back" (1 Peter 2:23)
Low impulsive reaction I	self-control	the same verse: "When he suffered, he didn't threaten"
Home h	peace	Philippians 4:7; Matthew 11:28–30

Only the words change, not the math. With faith on, every number (e, e', X, S, I, P) is identical; this is tested. The faith reading is offered as a way of speaking about the same quantities, not as a claim that the model measures anything spiritual.

4 What is proven, what is tested, what is modeled

Proposition 1 (Contraction still holds)**.** *Let T be the step map (2) with hard and process homes at 0. Then for every $e \in [0,1]^9$, $\|T(e) - h\| \le (1 - a_{\text{lo}})\|e - h\|$ with $a_{\text{lo}} = \min(a_0, a_{\max}, a_{\text{cap}}) > 0$. The Kai term does not change T, so the iteration converges to $e^* = h$, where $P = 1$.*

Proof. For a grounding mover, $T(e)_j - h_j = (1 - a)(e_j - h_j)$. For a hard or process mover, $T(e)_j = \text{clip}_{[0,1]}(e_j(1 - a - b\kappa_j))$ with $\kappa_j = -\sum_g \min(0, W_{gj})h_g \ge 0$, so $0 \le T(e)_j \le (1 - a)e_j$. Summing squares gives the bound. The Kai term enters only line (1): it takes a point of $[0,1]^9$ to another point of $[0,1]^9$ (clipped), raising only anger, shame and fear by at most Xw_j. The proof needs nothing else from the starting point, and the solve and recovery iterate T with empty text, where $X = 0$. □

Remark (If the pressure were applied at every step). Suppose the map were instead $T_X(e) = \mathrm{clip}_{[0,1]}(T(e) + p)$ with a constant $p = X(1 - S)w \geq 0$ on the hard movers. Then $|T_X(e)_j - h_j| \leq (1 - a)|e_j - h_j| + p_j$, so $d_{t+1} \leq (1 - a_{\mathrm{lo}})d_t + \|p\|$ and $\limsup d_t \leq \|p\|/a_{\mathrm{lo}}$. That is bounded, but it does *not* converge to h. This bound is loose (with the defaults $a_{\mathrm{lo}} = 0.12$). The app does not do this: outside hurt is applied once per reading.

Proposition 2 (The shield behaves as the story says)**.** *$S \in [0, 0.99]$ and $\partial S/\partial V = \prod_{U \neq V}(1 - U) \geq 0$ for each virtue V. With X fixed, $X(1-S)$ and the pushed anger fall as S rises, so u and $I = X(1-S)u$ are non-increasing in S; I is strictly decreasing when $X > 0$ and $u > 0$ (before rounding). With $X = 0$ nothing changes: no push, $I = 0$, and no card tile.*

Tested, not proven: the code matches (1)–(3) to the cent. The rounded iteration stops within $0.005/a$ of h. The detection separates hurt received from hurt given on the listed examples. Crisis wording returns only the 988/911 box. Everything is deterministic. The unit suite has 299 checks. **Modeled, not validated:** the movers' links, every weight and threshold, the cue lists, the virtue formulas, and the practice rules. No study with people has been run.

Credits and References

With gratitude to every researcher, clinician, translator and writer listed here, whose careful work MindFrame stands on. All credit and glory to God.

Every source used across MindFrame (the app, its README and its papers) was checked in October 2026. Journal articles were checked against Crossref (title, authors, year, volume, pages, DOI); books against Open Library or publisher and library catalogs; reports, preprints, archives and web articles at their source. 99 are verified; 2 are flagged and marked below. Films are credited to their studios and filmmakers and songs to their artists and songwriters; MindFrame only names them and quotes no lyrics. The works cited in this paper follow under *Works cited*.

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